Unit 5.1 Anti-derivatives

Note Title 2015/05/18

$$F \longrightarrow f \longrightarrow f'$$ (Differentiation process)

$F' = f$ and F is the anti-derivative of f

Also written $F = \int f(x)\,dx$ $\quad \left(f = \int f'(x)\,dx \right)$

So $F(x) = \int f(x)\,dx$ is the reverse process of differentiation.

Eg If $f(x) = 3x^2$ then $F(x) = x^3 + C$

If $f(x) = x^2$ then $F(x) = \frac{1}{3}x^3 + C$.

$$\boxed{\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C}$$

Eg If $f(x) = \cos x$ then $F(x) = \sin x$

If $f(x) = \underline{\cos(x^3) \cdot x^2}$ then $F(x) = \frac{1}{3}\sin(x^3)$

$$\frac{1}{\cancel{3}}\,\underline{\cos(x^3) \cdot \cancel{3}x^2}$$

Eg Find the antiderivative of:

1. $f(x) = e^x \sin e^x \Rightarrow F(x) = -\cos e^x + C.$

2. $f(x) = x(x^2+1)^4 \Rightarrow F(x) = \frac{1}{5 \times 2}(x^2+1)^5 + C$

$$= \frac{1}{10}(x^2+1)^5 + C$$